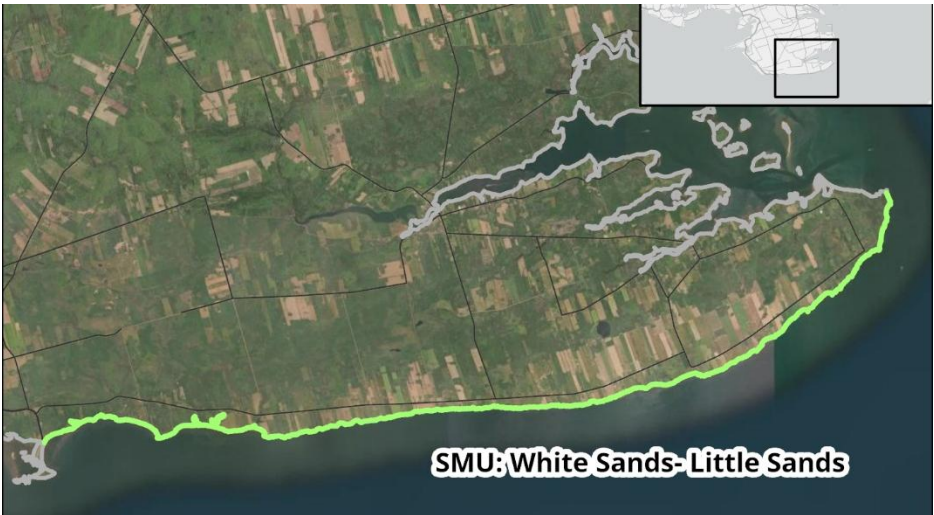


White Sands – Little Sands

Shoreline length: 38 km



Shore Types

White Sands – Little Sands is an exposed stretch of coast at the southeastern end of PEI with mostly cliffs and bluffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
6	16	0	66	5	0	7	0



Bluff at Wood Islands Provincial Park



Cliff at Cape Bear Lighthouse and Marconi Station



Cliffs along community of White Sands

Governance, Infrastructure, and Economy

Approximately 15% of this area is located within the Rural Municipality of Belfast; the remainder is unincorporated. Key infrastructure includes roads accessing beaches as well as roads extending along the

shoreline: Panting Shore Road at the beach; Shore Road (Route 4), east of the Buddhist Institute at Little Sands; MacFarlane Road (White Sands, at the beach); and Black Brook Road (Beach Point). Coastal economic drivers are limited. Local and legacy land use constraints include numerous small lots and unpaved coastal roads.

Natural Assets and Heritage

Key features include the Northumberland Provincial Park (playground, campground), the Cape Bear Lighthouse, the Marconi Station, and Guernsey Cove Beach. The shoreline includes critical habitat for bird species including the threatened Bank.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

The area has relatively low density of buildings and infrastructure within the coastal hazard zone. The small river mouths and coastal inlets to the west are vulnerable to increasing flood hazards. Erosion hazard is also highest along the exposed low plains.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

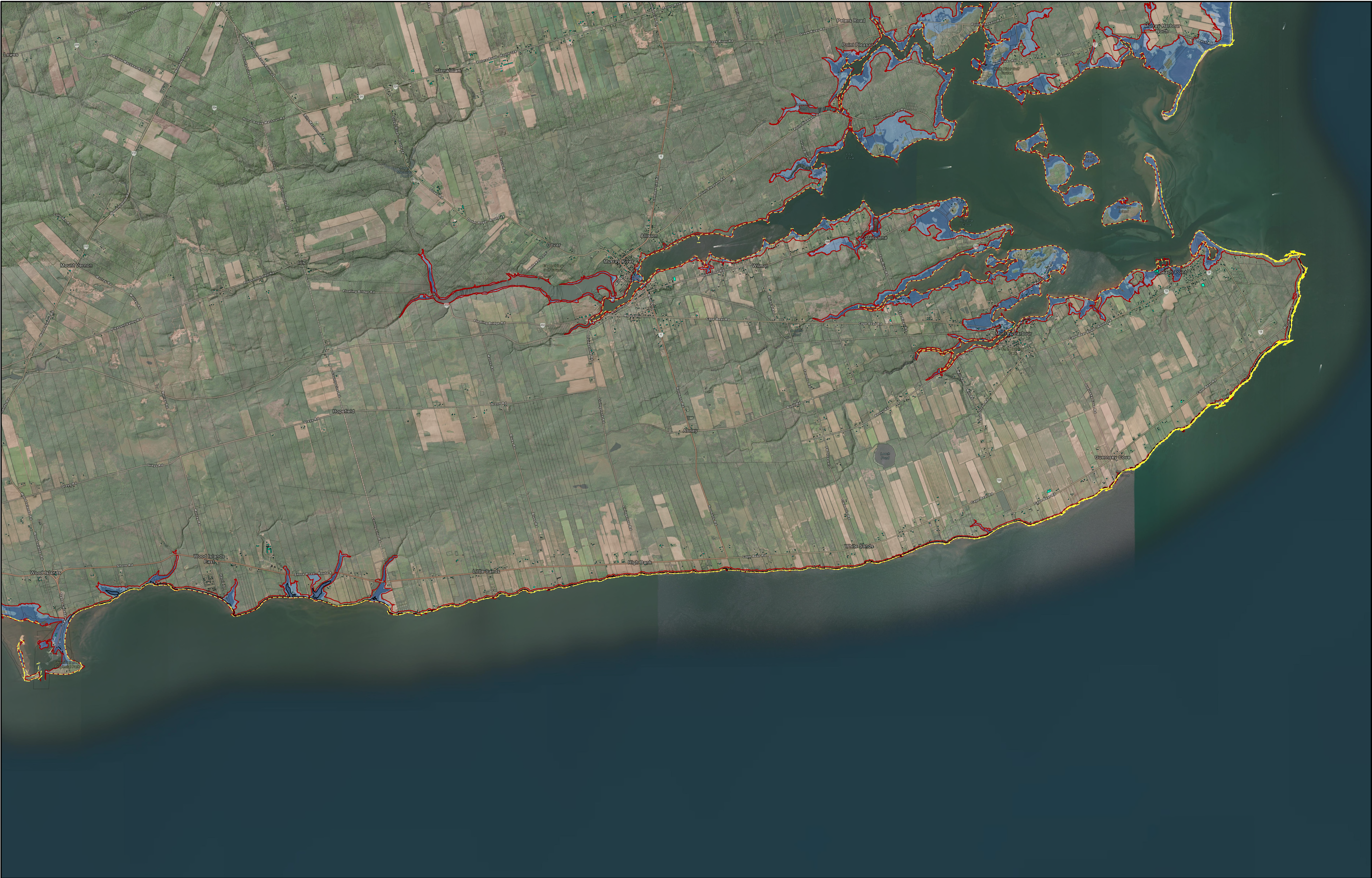
	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU White Sands- Little Sands				
Flood risk	2 [<1/km]	3 [<1/km]	3 [<1/km]	6 [<1/km]
Erosion risk	NA	9 [<1/km]	17 [<1/km]	32 [1/km]

Sediment transport - The sediment transport across the SMU is high, with eroding cliffs and bluffs feeding beaches typically from east to west.

Proposed SMP Strategies

No Active Intervention is proposed for the short-term, followed by Managed Realignment for the limited number of assets at risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including beaches and cliff habitat.		



LEGEND

- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

0 395 790 1,580
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



**COASTAL EROSION
AND FLOOD HAZARD**

**Shoreline Management Unit:
White Sands- Little Sands**

Prince Edward Island
Île-du-Prince-Édouard

CBCL